Change Detection in Remote Sensing Imagery: A Survey of Statistical, Machine Learning, and Deep Learning Methods

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Abstract

Change detection is a fundamental remote sensing task aimed at identifying surface modifications from multi-temporal imagery of the same geographic area, with applications in urban monitoring, agricultural survey, forest disturbance mapping, disaster assessment, and land cover analysis. The task is complicated by radiometric and atmospheric variability, co-registration errors, seasonal cycles, and sensor heterogeneity. Deep learning has progressively superseded traditional and classical machine learning approaches through its capacity for hierarchical feature extraction and end-to-end optimization. This survey comprehensively reviews change detection methods across three paradigms: traditional approaches (algebraic operators, transformation-based methods, probabilistic frameworks), classical machine learning (support vector machines, random forests, object-based image analysis), and deep learning architectures (fully convolutional networks, Siamese networks, attention mechanisms, transformers, state space models, diffusion-based approaches, and weakly supervised frameworks). We provide foundational background covering problem formulation, benchmark datasets, and evaluation metrics, alongside a structured taxonomy organized by methodological paradigm and supervision mode. A quantitative comparison on dominant benchmarks reveals the strengths and limitations of current methods. Key open challenges identified include the absence of a universal benchmark protocol, the gap between research performance and operational deployment, and the need for label-efficient learning strategies. This survey serves as a structured reference for the community and outlines promising directions for advancing change detection in remote sensing.

Keywords: change detection; remote sensing; deep learning; image classification; land cover monitoring; benchmark evaluation; comprehensive survey

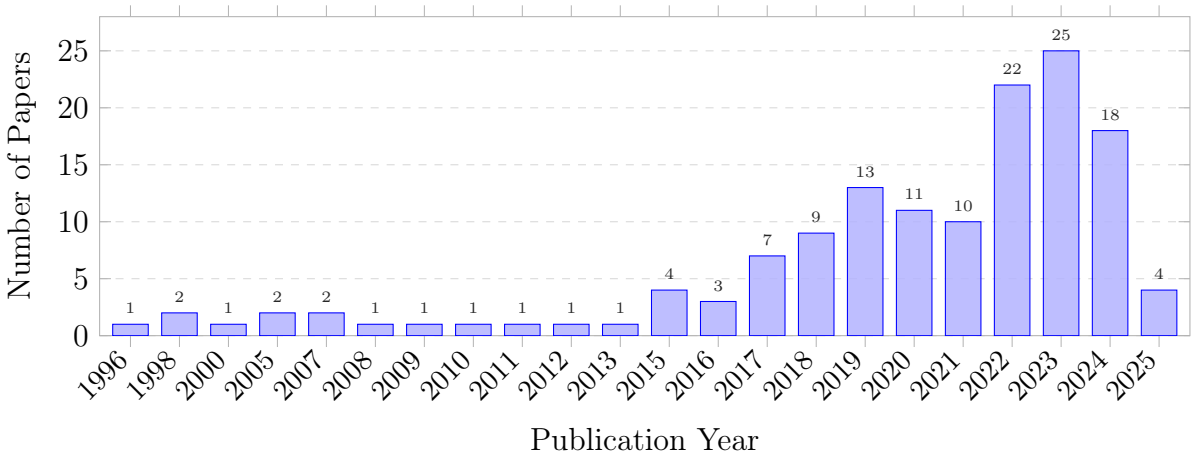

Figure 1: Temporal distribution of the 140 primary papers reviewed (1996–2025). An accelerating trend is evident from 2017, coinciding with the maturation of convolutional change detection benchmarks and the emergence of large-scale annotated datasets. The 2023 peak of 25 papers reflects the simultaneous emergence of transformer, Mamba, and weakly supervised architectures.

1 Introduction

Change detection in remote sensing refers to the process of identifying meaningful modifications in the state of the Earth's surface by analysing images of the same geographic area acquired at different time points [1,2]. The problem is conceptually straightforward: given two co-registered images acquired at times t_1 and t_2, identify pixels or regions that have undergone genuine semantic change. In practice, however, the task is complicated by geometric misregistration between acquisitions, atmospheric effects causing radiometric inconsistencies, seasonal phenological variability in vegetation, differential illumination, and the diversity of sensor modalities, spatial resolutions, and geographic contexts encountered across operational deployments.

The past decade has witnessed an exponential growth of Earth observation infrastructure and a corresponding surge in change detection research. Figure 1 illustrates the temporal distribution of the 140 papers comprising this review, revealing an acceleration commencing approximately in 2017, corresponding to the widespread adoption of convolutional neural networks and the emergence of the first large-scale annotated change detection benchmarks, and continuing through 2025, with 25 papers published in 2023 alone.

Figure 2 shows the methodological composition of the full corpus: deep learning accounts for 88 of the 140 papers (63%), classical statistical methods for 28 (20%), machine learning for 21 (15%), and manual or expert-based approaches for 3 (2%). Within the deep learning sub-corpus, supervised learning is dominant (74 papers, 84%), with weakly supervised methods, an emerging and practically important paradigm that operates under image-level rather than pixel-level annotation, comprising 11 papers (12.5%) and fully

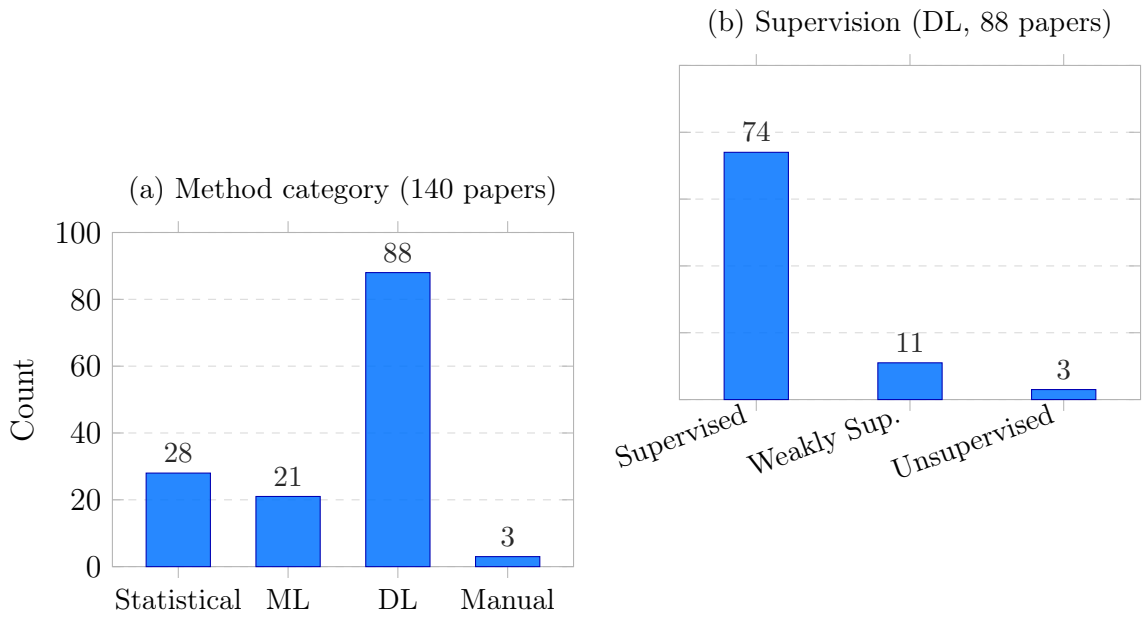

Figure 2: Distribution of the 140 primary papers by (a) methodological category and (b) supervision paradigm within the deep learning sub-corpus of 88 papers. Deep learning accounts for 63% of the corpus. Within the DL sub-corpus, supervised learning dominates (84%), with the weakly supervised paradigm representing an emerging and practically important research direction (11 papers, 12.5%).

unsupervised approaches the remaining 3 (3.5%). The architectural taxonomy in Figure 4 organises the surveyed methods across three epochs; the bi-temporal fusion strategy distribution in Figure 5 shows Siamese middle-fusion (69 of the DL papers, ≈59%) as the dominant paradigm.

This survey focuses on optical and multispectral remote sensing imagery, which constitutes the dominant modality in the reviewed corpus. Methods for synthetic aperture radar (SAR), hyperspectral, heterogeneous multi-modal, and three-dimensional point cloud data are discussed with representative references in Section 8. The survey spans terrestrial change detection across multiple application domains, as illustrated in Figure 3: general binary change detection (69 papers), building change (21), forest and vegetation disturbance (14), land cover/land use change (14), urban monitoring (8), and specialised domains including glacier, landslide, wetland, coral reef, and flood monitoring.

Several surveys have examined the change detection literature. Singh [1] established the foundational taxonomy of pre-digital and early digital methods. Lu *et al.* [2] provided the most comprehensive pre-deep-learning treatment. Coppin *et al.* [3] addressed ecosystem monitoring from an ecological perspective. Radke *et al.* [4] contributed a cross-disciplinary systems-level view. Cheng *et al.* [5] present the most recent systematic treatment of deep learning approaches through 2024. Lv *et al.* [6] reviewed heterogeneous remote sensing change detection with emphasis on cross-modal challenges. The present survey both complements and extends these works by: (i) spanning the full methodologi-

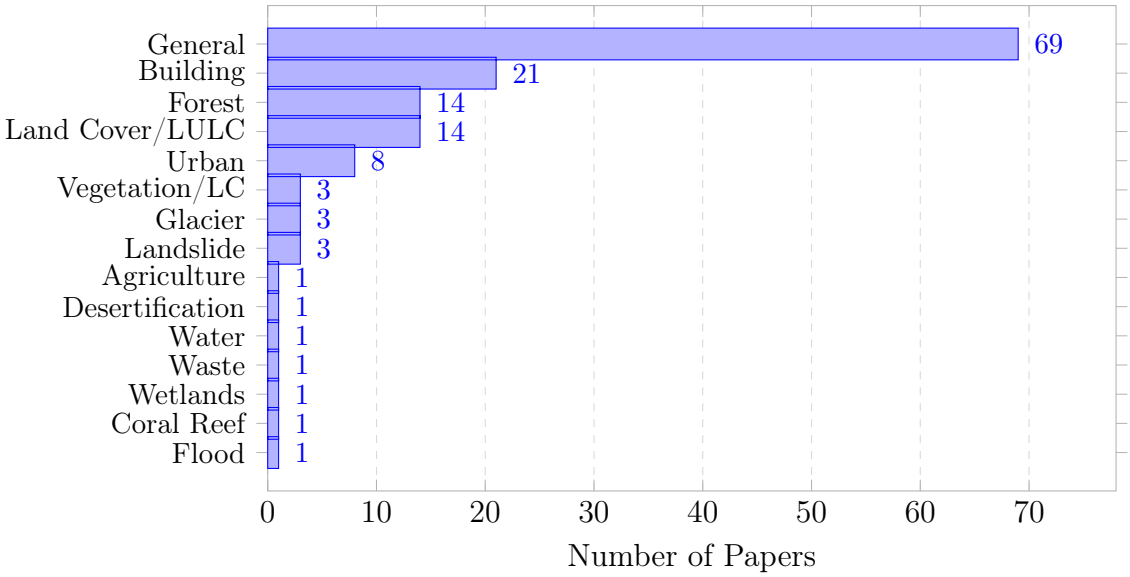

Figure 3: Distribution of the 140 primary papers by application domain. "General" denotes papers evaluated on standard binary change detection benchmarks without a specific application focus, constituting the largest category owing to the benchmark-centric evaluation context of the SOTA supervised and weakly supervised corpus papers. Building, forest, and land cover change collectively account for a substantial share of domain-specific papers.

cal arc from classical operators to Mamba- and diffusion-based architectures; (ii) treating the weakly supervised paradigm with the depth it merits given the 11-paper sub-corpus identified in this review; and (iii) combining insights from two complementary source datasets that together account for 140 non-redundant papers.

Section 2 introduces the formal problem definition and evaluation metrics. Section 3 describes the literature search protocol. Section 4 presents the principal benchmark datasets. Sections 5–7 review the three methodological categories. Section 8 addresses non-optical modalities. Section 9 presents the quantitative benchmark comparison. Section 10 identifies open challenges and future directions. Section 11 draws conclusions.

2 Background and Problem Formulation

2.1 Formal Definition

Let $\mathbf{X}_1 \in \mathbb{R}^{H\times W\times C}$ and $\mathbf{X}_2 \in \mathbb{R}^{H\times W\times C}$ denote two co-registered remote sensing images of the same geographic area acquired at times $t_1 < t_2$, where H and W are spatial dimensions and C is the number of spectral channels. The change detection problem is to produce a change map $\mathbf{M}$ via a detection function f:

$$\mathbf{M} = f(\mathrm{PR}(\mathbf{X}_1),\, \mathrm{PR}(\mathbf{X}_2))\,, \tag{1}$$

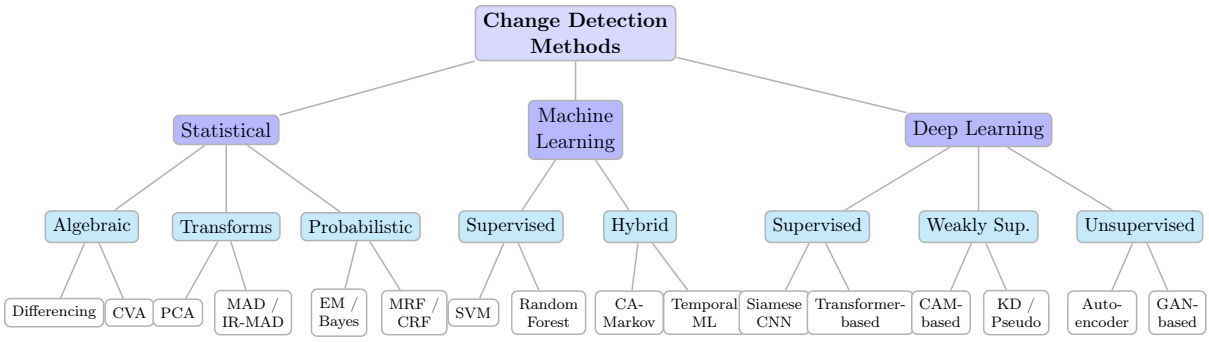

Figure 4: Taxonomic overview of the dominant change detection method families covered in this survey. The three top-level branches correspond to the three methodological epochs reviewed in Sections 5–7. For clarity, only the most representative leaf-level families are shown; additional architectures such as Mamba/SSM, diffusion models, patch-level weakly supervised methods, and object-based image analysis are discussed in the corresponding sections of the text.

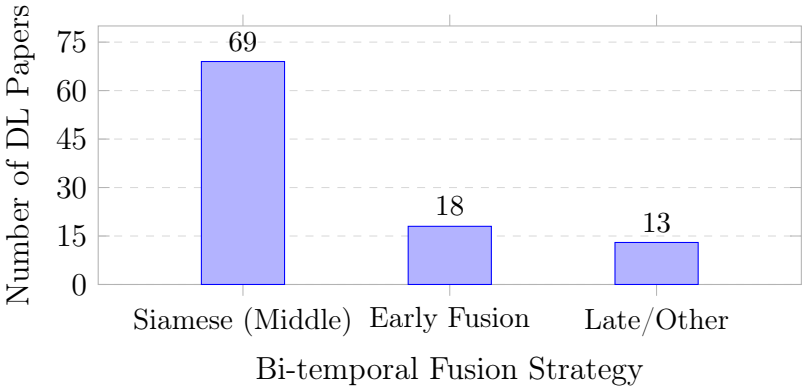

Figure 5: Distribution of bi-temporal fusion strategies among the 100 deep learning papers in the combined corpus. Siamese middle-fusion is the dominant paradigm (69 papers, 69%), reflecting the field's consensus that weight-shared feature extraction followed by explicit intermediate feature comparison outperforms early concatenation. The 18 early-fusion papers demonstrate that competitive performance is achievable with simpler single-stream architectures.

where PR(·) denotes image pre-processing and registration. Three bi-temporal fusion strategies are distinguished [5]: **early fusion** (concatenation along the channel dimension prior to feature extraction), **middle/Siamese fusion** (independent branch encoding with subsequent feature comparison), and **late fusion** (post-classification comparison of per-date label maps). In **binary change detection (BCD)**, $\mathbf{M} \in \{0,1\}^{H \times W}$. In **semantic change detection (SCD)**, $\mathbf{M}$ additionally characterises the land cover categories at t_1 and t_2, providing *from–to* transition information.

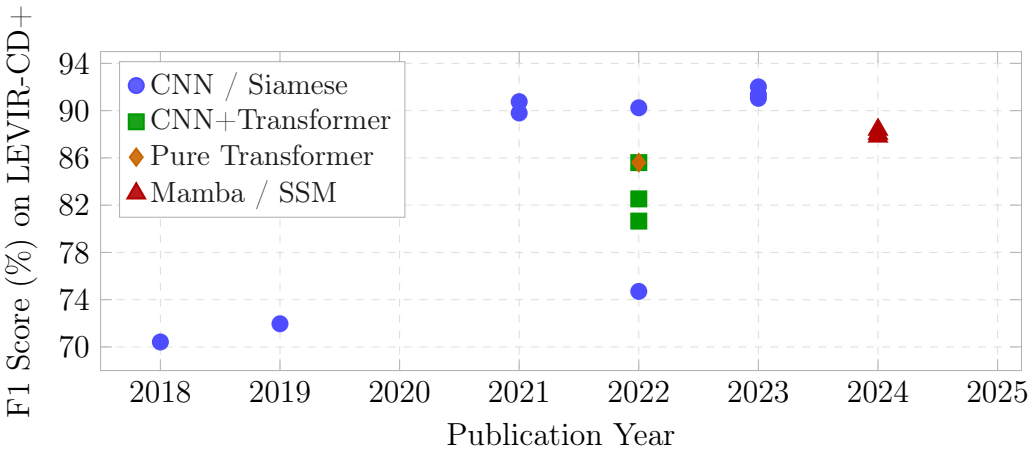

Figure 6: Performance progression on the LEVIR-CD+ benchmark by architectural paradigm (2018–2024). The dashed line indicates the approximate linear trend. The Mamba-based ChangeMamba achieves the current state of the art (F1 = 88.39%), while attention-augmented CNN Siamese methods (CGNet: F1 = 92.01%; USSFC-Net: F1 = 91.04%) demonstrate that global self-attention is not strictly necessary for competitive binary change detection.

2.2 Evaluation Metrics

Performance assessment relies on four fundamental quantities: true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). Standard metrics are:

$$F_1 = \frac{2\,\mathrm{TP}}{2\,\mathrm{TP} + \mathrm{FP} + \mathrm{FN}}, \qquad \mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP} + \mathrm{FN}}, \tag{2}$$

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}}, \qquad \kappa = \frac{p_o - p_e}{1 - p_e}. \tag{3}$$

For SCD, mean intersection-over-union (mIoU), separated kappa (Sek), and the composite F_{scd} score [7] are additionally employed. A critical observation, recurring throughout this survey, is that different papers report different metric subsets on different train/test splits, rendering direct cross-paper numerical comparison unreliable even within nominally shared benchmark datasets.

3 Literature Search Methodology

The reviewed literature was compiled from two complementary sources following a systematic inclusion protocol inspired by the PRISMA guidelines [8]. The first source, hereafter referred to as the *main corpus*, is a curated spreadsheet of 100 change detection papers spanning statistical, machine learning, and deep learning methods across diverse application domains, assembled through iterative expert review. The second source, the *SOTA corpus*, is a structured literature tracking dataset comprising three thematic sub-sheets: supervised deep learning, unsupervised deep learning, and weakly supervised deep learn-

ing, each paper assigned one of three status labels: READ, TO READ, or DISCARDED.

The inclusion rule applied to the SOTA corpus is: a paper is included if and only if its status is READ. Papers labelled TO READ are excluded as their analysis was not completed at the time of this survey's preparation; papers labelled DISCARDED are excluded following expert screening for relevance. This yielded 50 supervised, 1 unsupervised, and 11 weakly supervised SOTA papers, totalling 62 papers of which 15 overlap with the main corpus. After deduplication, 47 unique SOTA papers are added to the main corpus. An additional verification step removed 7 papers that are review articles or methodology references rather than primary change detection contributions, yielding a final corpus of **140 primary papers**. Figure 1 illustrates the temporal distribution of the combined corpus.

4 Datasets and Benchmarks

Table 1 summarises the major publicly available change detection benchmarks used across the 140-paper corpus. Within the DL papers, LEVIR-CD appears in 33 papers, CDD in 24, and WHU-CD in 18, confirming their status as the three primary evaluation standards for binary change detection.

LEVIR-CD [9]. Comprises 637 pairs of Google Earth images at 0.5 m GSD (1024×1024 pixels) covering American cities (2002–2018), with pixel-level binary building change annotations. The standard patch-level variant (LEVIR-CD+, 256×256 patches, 7120/1024/2048 train/val/test) is used in most contemporary benchmarking.

CDD [10]. Season-varying change detection dataset of 11 VHR image pairs (0.03–1 m/pixel) from Google Earth, capturing building construction, vegetation growth, and general land use modification. Its semantic diversity makes it a complementary benchmark to the building-focused LEVIR-CD and WHU-CD.

WHU-CD [11]. Aerial imagery at 0.3 m GSD over Christchurch, New Zealand before and after the 2011 earthquake, with binary building change annotations over approximately 20.5 km^2.

SYSU-CD [12]. 20,000 aerial image pairs at 0.5 m from Hong Kong (2007–2014) across five change scenario types: urban construction, suburban expansion, groundwork, vegetation change, road expansion, and sea construction.

DSIFN-CD [13]. 3940 pairs of 512×512 images from six Chinese cities (2 m GSD), providing a larger and more geographically diverse binary change detection benchmark

Table 1: Principal publicly available change detection benchmarks. GSD: ground sampling distance. Freq.: number of corpus papers using each dataset.

Dataset	Year	Modality	GSD	Pairs	Change classes	Freq.
LEVIR-CD [9]	2020	RGB (Google Earth)	0.5 m	637	1 (building)	33
CDD [10]	2018	RGB (Google Earth)	0.03–1 m	11	1 (binary)	24
WHU-CD [11]	2019	Aerial RGB	0.2–0.3 m	1 large pair	1 (building)	18
SYSU-CD [12]	2021	Aerial RGB	0.5 m	20,000	1 (5 scenarios)	9
GZ-CD [15]	2020	RGB (Google Earth)	0.55 m	19	1 (binary)	6
DSIFN-CD [13]	2020	RGB	2 m	3940	1 (binary)	3
SECOND [7]	2021	Aerial RGB	0.5–3 m	4,662	6 (semantic)	2
HRSCD [14]	2019	Aerial RGB	0.5 m	2 large pairs	5 (semantic)	2
BTCDD	2020	RGB (Google Earth)	–	5,281	1 (binary)	3
S2Looking	2021	Satellite	0.5–0.8 m	5,000	1 (building)	1
Air Change (AC)	2009	Aerial	1 m	3 pairs	1 (binary)	3
OSCD	2018	Multispectral (S-2)	10 m	24 pairs	1 (binary)	2

than LEVIR-CD.

HRSCD [14]. Two large-format (10,000×10,000) aerial image pairs from Bordeaux and Caen, France, annotated with five land cover classes. As explicitly noted in [14], the scarcity of such annotated datasets constitutes a principal bottleneck limiting the development of deep supervised semantic change detection methods.

SECOND [7]. 4,662 aerial image pairs (0.5–3 m) annotated with six land cover classes for from-to semantic change detection evaluation.

GZ-CD [15]. 19 season-varying VHR image pairs at 0.55 m GSD from Guangzhou, China (2006–2019), with binary building change annotations. The dataset exhibits substantial seasonal variation, making it a valuable complement to non-seasonal benchmarks.

Benchmark limitations that affect the interpretability of reported results include: (i) the three primary benchmarks disproportionately represent building change in East Asian and North American urban contexts; (ii) train/test split definitions and patch extraction strategies differ across papers even for the same nominal dataset; (iii) all existing benchmarks provide point-in-time binary annotations and do not support multi-temporal sequence evaluation; and (iv) no benchmark encompasses diverse geographic regions, multiple sensor modalities, and multiple change type categories simultaneously.

5 Traditional Statistical Methods

Statistical approaches, dominant from the 1970s through the early 2010s, typically follow a three-stage pipeline: (1) generation of a change indicator image through algebraic or transform-based operators applied to the co-registered image pair; (2) statistical analysis

and thresholding of the indicator image to produce a binary or multi-class change map; and (3) optional spatial post-processing for consistency enforcement.

5.1 Algebraic Difference and Index Methods

Image Differencing and Linear Operator Comparison. Collins and Woodcock [16] conducted the earliest systematic comparative evaluation of linear multitemporal change detection operators for forest mortality mapping, comparing multidate PCA, Gramm–Schmidt orthogonalisation, and the multitemporal Kauth–Thomas transformation on Landsat 5 TM data from the Lake Tahoe Basin (1988, 1991, 1994). Ground truth comprised quantitative basal-area loss from 25–27 forest stands per year. PCA and the Kauth–Thomas Wetness component were identified as the most reliable indicators (R^2 = 0.69–0.79), and notably, preprocessing beyond DN matching was found to provide no consistent benefit.

Spectral index differencing provides targeted sensitivity to specific biophysical processes. Liu *et al.* [17] proposed the NBRSWIR fire index to reduce false positives from water, smoke, and salt fields in post-fire burned area mapping. Applied to Landsat-8 OLI imagery of three wildfire events and followed by Otsu automatic thresholding, NBRSWIR achieved the highest Separation Index among compared methods, with overall accuracies of 98.93%, 98.57%, and 99.51% respectively. For vegetation monitoring, Meera Gandhi *et al.* [18] applied NDVI differencing to Landsat TM imagery of Vellore District (2001–2006), documenting a 19% increase in agricultural land and 23% decrease in open/barren land. Zhang *et al.* [19] extended spectral index approaches to forest health monitoring, proposing a spatiotemporal change detection method for pine wilt disease mapping from high-resolution imagery in complex mountainous landscapes.

The scale-heterogeneous LUCC detection method of Xing *et al.* [20] addressed the specific distortion introduced by resampling when comparing images of different spatial resolutions. By comparing scale-invariant MSER region features, Color-SIFT keypoints, spatial regression voting, and MRF smoothing directly across multi-resolution Montreal orthophotos (0.11–0.60 m, 2005–2012), total accuracies of 60.5–85.1% were achieved across six temporal pairs without resampling artefacts.

Change Vector Analysis. CVA [21,22] decomposes the per-pixel spectral change vector $\Delta\mathbf{x}_i = \mathbf{x}_{2,i} - \mathbf{x}_{1,i}$ into magnitude and direction components, enabling simultaneous binary detection and change type identification. Yang *et al.* [23] applied harmonic Landsat time-series modelling with CVA-based change characterisation for coastal tidal wetland monitoring across 17 tiles of the northeastern US (1984–2021), achieving land-cover classification accuracy of 95.8% and change detection accuracy of 99.8%, while importantly distinguishing between *cover change* and *condition change*. CVA also forms the initial

change-detection step in the CDMRF method of Lu *et al.* [24] for multi-sensor landslide mapping, where ICA-based variants improved upon CVA-CDMRF with kappa of 0.82–0.90 across four disaster events. Tsai *et al.* [25] applied NDVI filtering with CVA to Formosat-2 imagery for Typhoon Morakot landslide inventory, detecting 9,333 landslides over 22,590 ha with approximately 9% omission and 16% commission error.

5.2 Transformation-Based Methods

Principal Component Analysis. Applied to the stacked bi-temporal image, PCA concentrates change information into higher-order components. Al-Khudhairy *et al.* [26] employed PCA alongside image differencing, object-oriented eCognition classification, and mathematical morphology for structural damage assessment from Ikonos data following conflict events in Jenin and Brest (FYROM). Standardised PCA achieved 12% omission and only 3% commission on the Brest dataset. The unsupervised PCA + k-means pipeline [27] remains among the most widely cited baseline methods for unsupervised change detection, providing a transparent, parameter-light reference point against which more complex methods are evaluated.

Multivariate Alteration Detection and IR-MAD. Nielsen's iteratively reweighted MAD transform [28, 29] applies canonical correlation analysis in successive iterations, down-weighting likely changed pixels to progressively refine the change-free correlation structure. IR-MAD is affine-invariant, making it robust to differential radiometric calibration between acquisitions. Within the current corpus, IR-MAD is exploited as a pre-detection strategy in a hybrid unsupervised transformer-based framework [30], where IR-MAD and compressed CVA jointly generate pseudo-training samples from VHR imagery of Guangzhou and Nanjing without manual annotation, subsequently initialising a transformer network that surpasses standalone IR-MAD performance.

Time-Series Decomposition. For vegetation and land cover monitoring, spectral time-series decomposition enables detection of both abrupt and gradual changes. Ghaderpour and Vujadinovic [31] proposed JUST (Jumps Upon Spectrum and Trend), a weighted least-squares method operating on unequally spaced, noisy satellite time series to simultaneously estimate jump location, direction, and magnitude. Applied to MODIS NDVI and Landsat-8 EVI time series, JUST correctly detected 94% of wildfire-related negative-magnitude jumps within one month of occurrence versus 86% for OLS-based methods. Ben Abbes *et al.* [32] systematically compared STL, BFAST, and MRA-WT decomposition methods on simulated NDVI series, finding average trend RMSE of 0.015 for BFAST versus 0.037 for MRA-WT and demonstrating that method selection should reflect the expected change dynamics of the target ecosystem. Winsvold *et al.* [33] applied Red/SWIR band-ratio time-series stacking to Landsat imagery for regional glacier mapping across

three geographically diverse glacier systems, demonstrating that temporal stacks provide more robust delineation than single optimal-date selections ($R^2 = 0.27$–0.39 for seasonal ratio elevation correlations). Hughes *et al.* [34] introduced VeRDET for patch-based Landsat time-series forest disturbance detection using total variation regularisation, classifying temporal segments as disturbed, stable, or regenerating (overall error ≈23%, with omission varying from 22% for wildfire to 78% for physiological stress).

5.3 Probabilistic and MRF-Based Methods

Markov Random Field regularisation reduces false positive rates by modelling spatial label dependencies within the change map. The foundational probabilistic framework for unsupervised change detection was established by Bruzzone and Fernández Prieto [35], who formulated the problem as binary hypothesis testing over the difference image using a Gaussian mixture model with EM parameter estimation for automatic threshold selection. Lu *et al.* [24] improved CDMRF by substituting CVA with δNDVI, PCA, and ICA alternatives for the change-detection image generation stage, achieving kappa of 0.90 for the Jiuzhaigou earthquake scenario and demonstrating multi-sensor capability (pre-event Landsat-8 combined with post-event Sentinel-2). Byun *et al.* [36] proposed an unsupervised flood extent extraction method exploiting spectral distortion during cross-fusion of bi-temporal KOMPSAT-2 VHR imagery: the pre-flood multispectral image is fused with the normalised post-flood NIR image, local distortion is measured via UIQI, and EM thresholding separates flooded from non-flooded pixels. An OA of 75.04%, comparable to supervised SVM and substantially outperforming CVA and MAD, was achieved while requiring no training data. Luo *et al.* [37] applied Dempster–Shafer evidence theory to multitemporal VHR imagery for urban change detection, fusing spectral, textural, and structural evidence sources under an uncertainty-aware framework.

5.4 Post-Classification Comparison

Post-classification comparison (PCC) independently classifies each date and identifies change as label discrepancies [2]. PCC constitutes the dominant approach for long-term land use/land cover (LULC) change analysis in the corpus. Rawat *et al.* [38] documented built-up expansion of 8.88% in Ramnagar town (1990–2010, Landsat TM, 30 m). Rawat and Kumar [39] reported vegetation increase and barren land decrease in Hawalbagh block (OA 90.29–92.13%, kappa 0.82–0.91). Tewabe and Fentahun [40] quantified 13–15% cultivated land increase in the Lake Tana Basin, Ethiopia (1986–2018, Landsat TM 30 m, OA improving from 84.21% to 90.36%). Hassan *et al.* [41] documented dramatic urban expansion of Islamabad from 16,281 ha (18.09%) to 51,039 ha (56.73%) using a mixed Landsat 5 TM / SPOT 5 pair. Tran *et al.* [42] reconstructed LULC transitions across nearly 40 years of the Mekong Delta (1973–2011, Landsat/SPOT, OA 81–89%), documenting the

conversion from rice cultivation to aquaculture.

CA-Markov frameworks extend PCC by training transition probability matrices on detected historical changes to simulate future land cover states. Koko *et al.* [43] applied this approach to Zaria city (1990–2020, Landsat, OA 84–88%), projecting future built-up expansion for 2035 and 2050. Belay *et al.* [44] contrasted business-as-usual versus governance scenarios in the Gumara Watershed, Ethiopia (1985–2019, Landsat, OA 91–94%), projecting cultivated land expansion to 89% under BAU versus stabilisation under the governance pathway. Lyons *et al.* [45] combined empirical bathymetry retrieval with maximum likelihood seagrass classification on bi-temporal QuickBird-2 imagery (2.4 m, 2004–2007, Moreton Bay), achieving classification accuracy of 57–95% across map units. Mas *et al.* [46] demonstrated a semi-automatic map-updating workflow combining automatic SPOT-5 segmentation, discrepancy detection, auxiliary C5 decision tree, and visual interpretation for Michoacán, Mexico, achieving OA = 83.3% ± 3.1%.

6 Machine Learning Methods

Machine learning methods introduce discriminative classifiers trained from labelled examples, learning change-discriminative functions from feature vectors derived from bi-temporal image pairs. An early and influential application of SVMs to change detection was proposed by Bovolo *et al.* [47], who demonstrated SVM classification of the difference image for binary change detection from multitemporal remote sensing data.

6.1 Support Vector Machines

Tan *et al.* [48] compared RF versus SVM under OBIA and PBIA paradigms on Sentinel-2A imagery (20 m) for post-wildfire land cover change in Serra da Estrela Natural Park, Portugal. RF-OBIA achieved the best OA (≈0.99, kappa 0.98–0.99), with NBR and ΔNBR identified as the most discriminative variables through RF importance analysis. An important practical observation was the divergence between numerical metrics and visual plausibility: SVM-PBIA maps matched reference data better visually in some heterogeneous burned areas despite lower overall accuracy scores. Azzouzi *et al.* [49] applied object-based SVM to five Landsat-5 TM scenes (1986–2011) for desertification monitoring in Biskra, Algeria, achieving OA of 84.89–93.95% across classification years, with sand area gain of 20.82% as the dominant change type. Atef *et al.* [50] compared SVM, RF, and MLH on Landsat data for El-Fayoum Governorate, Egypt (2000–2020), finding SVM best (OA 97.03%, kappa 0.96) and documenting urban land expansion from 3.43% to 5.99%. Dammalage and Jayasinghe [51] applied SVM to Landsat imagery of Colombo District (OA 99.04%), directly linking the growth of urban area from 10% to 36% between 1989 and 2016 to a corresponding increase in flood inundation extent from

63.0 km^2 to 74.7 km^2 despite lower total rainfall in 2016.

Hou *et al.* [52] proposed a semi-supervised building change detection framework combining EMAP spectral-spatial features, SLIC superpixel segmentation, saliency-guided pseudo-label generation, and Random Forest classification, achieving kappa of 0.52–0.75 across QuickBird and Google Earth image pairs while requiring no pixel-level manual annotation. This approach constitutes a conceptual bridge between unsupervised classical analysis and label-intensive deep supervised learning.

6.2 Random Forests and Ensemble Methods

Rash *et al.* [53] compared SVM, RF, ANN, KNN, and XGBoost for LULC change detection in the Kurdistan Region of Iraq using Landsat data (1991–2021), finding RF consistently best (kappa 0.93–0.97), with agricultural land increase of 13.54% (+480 km^2) and rangeland decrease of 11.33% over 30 years. Hu and Hu [54] applied annual RF classification via GEE to Landsat data across Central Asia (2001–2017), achieving overall accuracy of 0.90 ± 0.01 and identifying precipitation and drought as dominant drivers of natural vegetation change through multiple stepwise regression. Scharsich *et al.* [55] addressed missing historical ground truth by transferring recent RF class labels backward via CVA-identified stable pixels, achieving cross-validation consistency of 94.11–99.60% across three Landsat dates. Seydi *et al.* [56] applied Harris hawks optimisation (HHO) feature selection across 121 candidate features from Sentinel-2 MSI (10 m) for Australian wildfire damage assessment, with RF achieving OA = 91.02%, kappa = 0.82 using only HHO-selected features.

Seo and Kim [57] proposed RF regression for radiometric and phenological normalisation as a preprocessing step for change detection, achieving R^2 of 0.90–0.92 per band and downstream change detection OA of 95.30% with SLIC-based denoising, substantially outperforming linear normalisation baselines. In a subsequent study, Seo *et al.* [58] extended the RF normalisation framework to generate radiometrically and phenologically consistent reference images for improved multi-date comparison. Chen *et al.* [59] demonstrated RF-based one-class classification for earthquake-induced landslide susceptibility mapping using an incomplete inventory, addressing the practical constraint that post-disaster reference data are often spatially incomplete.

6.3 Object-Based Image Analysis

Wang *et al.* [60] demonstrated urban change detection from bi-temporal QuickBird (0.61 m, Xuzhou) and Ziyuan-3/GF-1 (2 m, Jiangyin) images using spectral, shape, and GLCM texture features combined via KNN/SVM/ELM/RF ensemble voting, achieving OA = 99.21% and kappa = 0.9835. Tan *et al.* [61] simultaneously addressed salt-and-pepper noise, segmentation-scale sensitivity, and single-classifier instability through multi-scale

CVA-guided sample generation, SVM+KNN+Extra-Trees classification, and Dempster-Shafer fusion, achieving OA = 0.9698 on ZY-3 imagery (5.8 m, Xuzhou 2012–2013). Zhou *et al.* [62] evaluated object-based versus pixel-based paradigms for coral reef change detection using QuickBird and WorldView-2 imagery (0.5–2.4 m) of four Spratly Islands sites, finding OBCD achieving average OA exceeding 90%, approximately 20% above the PBCD average of 69.72%.

6.4 Temporal Modelling and Hybrid Approaches

Hu *et al.* [63] combined CART classification with CVAPS posterior-probability change analysis and NDVI timing analysis for large-scale land map updating over 6.86 million km^2 of western China via GEE, demonstrating that NDVI timing analysis increased change-detection accuracy from 82.00% to 88.20%. Francini *et al.* [64] applied a reinforcement-learning-inspired temporal decision algorithm to PlanetScope RGB imagery (3 m) for near-real-time forest clearcut detection in Tuscany, achieving user's accuracy = 86% and producer's accuracy = 92% with detection latency of weeks rather than months. Roy and Inamdar [65] demonstrated that multi-sensor radiometric calibration is the key enabler for reliable long-term change analysis, achieving OA = 91.36% (kappa 0.91) across six Landsat dates (1972–2016) over the Shivna River Basin. Viana *et al.* [66] applied time-weighted dynamic time warping (TWDTW) to long-term satellite image time series for LULC change detection, demonstrating the value of temporally adaptive similarity measures for phenologically complex landscapes. Patel *et al.* [67] combined GIS and Google Earth Engine for spatiotemporal LULC analysis of Ahmedabad city, illustrating the growing role of cloud-based platforms in enabling large-scale change detection workflows.

7 Deep Learning Methods

Deep learning methods have transformed change detection since approximately 2016, driven by the convergence of large annotated benchmarks, increased computational resources, and successive architectural innovations. Figure 4 provides the full architectural taxonomy; Figure 5 shows the distribution of bi-temporal fusion strategies; Figure 6 traces the performance trajectory on LEVIR-CD+.

7.1 Fully Convolutional Networks and Foundational Siamese Designs

Caye Daudt *et al.* [68] introduced three FCN architectures (FC-EF early fusion, FC-Siam-conc and FC-Siam-diff Siamese middle fusion), all built on U-Net with skip connections. The Siamese middle-fusion configurations substantially outperformed early fusion, es-

tablishing the dual-branch weight-sharing paradigm as the dominant baseline. This is reflected in Figure 5: 69 of the DL papers in the combined corpus employ Siamese middle fusion.

Peng *et al.* [69] extended this to UNet++ with nested dense skip connections and multi-scale output fusion (MSOF), using a combined balanced binary cross-entropy and dice coefficient loss ($\lambda = 0.5$), achieving F1 = 0.8756 and OA = 0.9673 on the CDD predecessor dataset. Fang *et al.* [70] introduced SNUNet-CD combining NestedUNet with the Ensemble Channel Attention Module (ECAM), preserving shallow-layer high-resolution features to address localisation loss in deep encoders. Mou *et al.* [71] introduced ReCNN for joint spectral-spatial-temporal modelling on Landsat ETM+ (30 m), achieving OA = 98.73% and kappa = 0.9705 (binary) and demonstrating that recurrent temporal modelling substantially improves detection of gradual changes relative to single bi-temporal comparison. de Bem *et al.* [72] evaluated U-Net, SharpMask, and ResUNet for annual Amazon deforestation mapping on Landsat-8 OLI (30 m), with ResUNet achieving F1 = 0.9432–0.9465 and outperforming RF and MLP baselines by ≈0.05 in all metrics.

The deeply supervised image fusion network for DSIFN-CD [13] introduced a six-branch architecture integrating deep supervision at multiple decoder levels, evaluated on the six-city DSIFN-CD benchmark it simultaneously introduced. Shi *et al.* [12] presented the SYSU-CD dataset alongside a deeply supervised attention metric-based network achieving competitive binary change detection results. Peng *et al.* [15] proposed SemiCDNet, a semisupervised GAN-based architecture exploiting both labelled and unlabelled data through dual discriminators enforcing segmentation and entropy map consistency, introducing the GZ-CD dataset of 19 season-varying VHR image pairs from Guangzhou (0.55 m, 2006–2019). Bai *et al.* [73] introduced an edge-guided recurrent CNN that explicitly models building boundary information across temporal sequences, improving the delineation of fine-grained structural changes in multitemporal VHR imagery.

7.2 Attention-Augmented Siamese Networks

Chen and Shi [74] introduced DASNet to address pseudo-change sensitivity through dual spatial and channel attention augmenting a VGG-16 Siamese backbone, combined with a Weighted Double-Margin Contrastive (WDMC) loss for class imbalance. On LEVIR-CD: Precision = 0.8967, Recall = 0.9136, F1 = 0.8980, kappa = 0.8829. Song and Jiang [75] contributed the important demonstration that *early-fusion* single-stream architectures can match Siamese middle-fusion performance: AGCDetNet with DilatedResNet50 backbone, CG-ASPP, SPAM, and CIFU achieved F1 = 0.9076 (IoU = 0.8309) on LEVIR-CD and F1 = 0.9654 on the Season-Varying dataset.

Song *et al.* [76] introduced SUACDNet with three explicit interaction branches (dif-

ference, similarity, global) between Siamese streams via MCRM, SAM, FFM, and CG-CAM modules, achieving F1 = 0.9028 on LEVIR-CD. Ding *et al.* [77] proposed DSA-Net with the crucial architectural variant of *independent weights* in the two encoder branches (pseudo-Siamese), allowing per-date appearance specialisation: P = 0.8974, R = 0.8812, F1 = 0.8892 on LEVIR-CD (128×128 patches). Wang *et al.* [78] proposed ADS-Net with multilevel attention-guided fusion and four-branch deep supervision: F1 = 0.8980, kappa = 0.8829 on LEVIR-CD, and F1 = 0.8979 on SVCD.

Han *et al.* [79] introduced HANet with progressive foreground-balanced sampling (PFBS) to address extreme class imbalance (4.26–4.65% changed pixels in WHU-CD and LEVIR-CD). The hierarchical axial (row + column) attention with a hybrid weighted cross-entropy + dice loss achieves WHU-CD: F1 = 88.16, IoU = 78.82; LEVIR-CD: F1 = 90.28, IoU = 82.27, using only 3.03M parameters. Han *et al.* [80] introduced CGNet where a deep change guiding map derived from semantic features explicitly guides subsequent multiscale feature fusion through three CGM modules, targeting broken edges and internal holes in change maps: LEVIR-CD F1 = 92.01, IoU = 85.21.

Lei *et al.* [81] proposed DESSN with Difference Enhancement (DE), Spatial-Spectral Nonlocal (SSN), and asymmetric Ghost convolution modules, achieving LEVIR-CD F1 = 91.36 and CDD F1 = 94.53 with reduced computational cost. Lei *et al.* [82] introduced USSFC-Net, an ultralightweight pseudo-Siamese design using only 1.52M parameters and 4.86G FLOPs, with MSDConv multiscale extraction and SSFC spatial-spectral attention: LEVIR-CD F1 = 91.04, CDD F1 = 94.74.

The HFA-Net architecture [83] introduced a High Frequency Attention Block (HFAB) for building boundary detection, decomposing features into high- and low-frequency components and applying attention selectively to the high-frequency component. The AER-Net [84] paper simultaneously introduced the HRCUS-CD dataset of high-resolution urban change detection images, expanding benchmark diversity beyond Google Earth sources.

ADHR-CDNet [85] combined a high-resolution backbone with a Differential Pyramid Module (DPM) and Multiscale Spatial Feature Attention Module (MSSAM) for simultaneous multi-scale feature extraction and small-object detection. Wang *et al.* [86] applied a deep Siamese network with Hybrid Convolutional Feature Extraction Module (HCFEM) incorporating dilated multi-branch convolution to multi-sensor change detection (ZY-3/GF-2, 5.8 m), with object-based post-processing achieving OA = 0.9715, kappa = 0.7801. Xu *et al.* [87] proposed a hybrid framework combining RF pixel-level classification with improved U-Net semantic segmentation, achieving OA = 92.72% versus 89.39% for PCA alone on a 2.5 m dataset.

The LightCDNet architecture [88] addressed efficiency through a lightweight design retaining competitive accuracy on VHR images. LSNET [89] proposed an extremely lightweight Siamese network evaluated on the Air Change, LEVIR-CD, and WHU-CD benchmarks, establishing the lightest competitive baseline for resource-constrained de-

ployment. TINYCD [90] introduced the Mix and Attention Mask Block (MAMB) combining space-time domain feature mixing with MLP-based semantic attention, achieving performance competitive with substantially larger architectures on LEVIR-CD and WHU-CD.

WRICNet [91] proposed a weighted rich-scale inception coder for multi-scale feature extraction, evaluated on LEVIR-CD, CDD, and GZ-CD. ISNet [92] introduced margin maximisation to improve semantic boundary separability in high-resolution images. Li *et al.* [93] proposed a lightweight architecture with progressive feature aggregation and supervised attention for efficient change detection. More recent attention-based methods continue to refine multiscale feature fusion: Ren *et al.* [94] introduced dual-attention-guided multiscale feature aggregation combining spatial and channel attention at multiple encoder levels; Zhan *et al.* [95] proposed AMFNet with attention-guided multi-scale fusion for bi-temporal imagery; Jiang *et al.* [96] combined difference and attention mechanisms under multi-scale feature fusion for high-resolution city change detection; and Zhang *et al.* [97] introduced B2CNet, a progressive boundary-to-center refinement network that explicitly models change boundary localisation before interior region filling.

7.3 CNN-Transformer Hybrid Architectures

Chen *et al.* [98] established BIT as a landmark hybrid architecture, using ResNet feature extraction followed by a compact Transformer for spatiotemporal context modelling: LEVIR-CD+ F1 = 82.53%, OA = 98.60%; WHU-CD F1 = 90.04%. Wang *et al.* [99] proposed MTCNet combining ResNet18 with SAM, a four-scale transformer, and CAM: LEVIR-CD OA = 97.02, F1 = 90.24. Song *et al.* [100] introduced ACABFNet with parallel CNN and Transformer branches connected via axial cross-attention, using early fusion (concatenated bi-temporal input): BTCDD F1 = 82.39, CDD and LEVIR-CD evaluated.

The Dual-Branch Multilevel Intertemporal Network [101] addresses unbalanced variations under seasonal turnover and illumination change through unified self-attention (within-date) and cross-attention (across dates) mechanisms, evaluated on LEVIR-CD, WHU-CD, GZ-CD, and SYSU-CD. ChangeFormer [102] presented a hierarchical Transformer-based Siamese network with overlapping patch embedding for rich feature representation: LEVIR-CD F1 = 80.65%, WHU-CD F1 = 86.88%. ConvTransNet [101] addressed small changed area limitations in existing cascaded CNN-Transformer designs through a parallel ConvTrans block generating multi-scale global-local features simultaneously: LEVIR-CD and CDD. VcT [103] emphasised the role of unchanged background context information in change feature learning. The Asymmetric Cross-Attention Hierarchical Network [104] introduced AMCA bidirectional information exchange between CNN and Transformer components, reducing computational redundancy.

WNet [105] proposed a W-shaped hierarchical network for multi-scale feature ex-

traction in change detection. TransY-Net [106] introduced a fully transformer network with Y-shaped design for comprehensive change detection. The Triple Change Detection Network [107] exploited joint multifrequency and full-scale Swin-Transformer processing. MISGNet [108] introduced multilevel intertemporal semantic guidance for improved temporal correlation modelling.

ICIF-Net [109] introduced intra-scale cross-interaction at each encoder scale, enabling bidirectional temporal information flow at every resolution level, evaluated on LEVIR-CD, WHU-CD, GZ-CD, and SYSU-CD. MSFF-CDNet [110] proposed multiscale feature fusion for bi-temporal high-resolution change detection. Chen *et al.* [111] proposed Siamese_AUNet combining ASPP and Feature Attention Module (FAM), achieving F1 = 0.8557 on LEVIR, evaluated also on WHU-CD and SZTAKI Air Change. MSTDSNet-CD [112] combined Multiscale Swin Transformer with deep supervision, targeting urban LULC change in VHR images. Peng *et al.* [113] introduced SCDNet for semantic change detection with Siamese UNet, pretrained ResNet34, MAC attention, and three auxiliary classifiers: Sensetime mIoU = 0.7306, HRSCD mIoU = 0.6950. Sefrin *et al.* [114] applied FCN + ConvLSTM to 13-band Sentinel-2 data (10 m) over Saxony, Germany, demonstrating that temporal sequence modelling improves land cover classification kappa by 3–5 percentage points versus monotemporal FCN. Yin *et al.* [115] applied U-Net-LSTM for lake boundary prediction from annual Google Earth imagery of Lake Urmia (1996–2014), achieving ACC = 0.8943, DICE = 0.9608. The FTN [116] demonstrated a fully transformer network for change detection, evaluated on LEVIR-CD, CDD, and GZ-CD. Chen *et al.* [117] presented FD-MCD for unsupervised multimodal change detection using Fourier-domain structural relationship analysis, achieving best OA, F1, and kappa on all five SAR-optical datasets: River F1 = 0.7216, Shuguang F1 = 0.8136, Farmland F1 = 0.8931. Sublime and Kalinicheva [118] proposed an unsupervised deep learning pipeline for post-disaster mapping on ASTER (15 m) Tohoku tsunami imagery, using joint convolutional autoencoder cross-reconstruction: kappa = 0.51 versus 0.38 for the RBM baseline. Liu *et al.* [119] applied a CNN-Transformer network with multiscale context aggregation to fine-grained cropland change detection, demonstrating that transformer-enhanced contextual reasoning improves discrimination of subtle agricultural transitions. Wang *et al.* [120] proposed a dual-branch network combining transformer global modelling with CNN local feature extraction for complementary bi-temporal representation. Long *et al.* [121] addressed semantic change detection through a hierarchical semantic graph interaction network that models inter-class relationships from high-resolution imagery.

7.4 Pure Transformer and Full-Scale Architectures

Zhang *et al.* [122] introduced SwinSUNet, the first pure Swin Transformer architecture for change detection, with hierarchical shifted-window self-attention providing global con-

text: LEVIR-CD+ F1 = 85.60%, OA = 98.92%; WHU-CD F1 = 93.04%, IoU = 87.00%. Zheng *et al.* [123] implemented ChangeMask for semantic change detection with two inductive biases (semantic-change causality and temporal symmetry), jointly predicting binary change maps and per-date semantic segmentation: SECOND and Hi-UCD benchmarks. Lu *et al.* [124] proposed a bitemporal attention transformer that jointly addresses building change detection and building damage assessment within a unified architecture, extending the pure transformer paradigm to multi-task change analysis.

7.5 Mamba and State Space Model Architectures

Chen *et al.* [125] introduced ChangeMamba, applying the selective state space model (SSM) Mamba architecture to change detection. The linear $\mathcal{O}(n)$ computational complexity of SSM provides a fundamental advantage over the quadratic $\mathcal{O}(n^2)$ complexity of transformer self-attention. The VMamba encoder employs a Cross-Scan Module (CSM) unfolding 2D patches into four directional 1D sequences, and three spatio-temporal relationship modelling mechanisms combine global Mamba representations with explicit bi-temporal comparison. Performance: MambaBCD-Base on LEVIR-CD+ F1 = 88.39%, OA = 99.06%, IoU = 79.20%; WHU-CD F1 = 94.19%, OA = 99.58%, IoU = 89.02%, representing the current state of the art on both benchmarks. Designated ESI Highly Cited and Hot Paper, this work establishes Mamba as the leading architectural paradigm for change detection at time of writing.

7.6 Diffusion Model Approaches

Bandara *et al.* [126] pre-trained a DDPM on large collections of unlabelled remote sensing images, then extracted multi-timestep feature representations from the frozen denoising network as inputs to a lightweight hierarchical change classifier. Features at early timesteps encode fine-grained spatial detail while later timesteps provide semantic abstraction, constituting a multi-scale feature pyramid learned without change annotations. DDPM-CD significantly outperforms prior self-supervised change detection methods on LEVIR-CD, WHU-CD, DSIFN-CD, and CDD. Wen *et al.* [127] proposed a supervised conditional DDPM framework conditioning a diffusion-based noise predictor on the bitemporal image pair through a CD Conditioner module and Frequency Cross Transformer.

7.7 Weakly Supervised Change Detection

Weakly supervised change detection (WSCD) operates under image-level binary annotation rather than pixel-level labels, providing a practically important middle ground between annotation-free unsupervised methods and annotation-intensive fully supervised

training. Eleven WSCD papers with status READ were identified in the SOTA weakly supervised sheet.

Zheng *et al.* [128] introduced the foundational WSCD framework using a weakly supervised convolutional network for change segmentation and classification from image-level labels (2020). Wang *et al.* [129] developed CS-WSCDNet integrating class activation mapping (CAM) with the Segment Anything Model (SAM) [130] for zero-shot pseudo-label refinement, demonstrating that foundation model integration substantially improves weakly supervised pseudo-label boundary quality. Zhao *et al.* [131] addressed the instance-lumping failure mode via Plug-and-Play DISep: a three-step iterative procedure of high-pass CAM localisation, connectivity-based instance retrieval, and intra-instance separation loss to better delineate individual changed objects. Li *et al.* [132] identified change-missing and change-fabricating as two distinct WSCD failure modes, addressing them via dilated-prior decoders and label-gated constraints. Lu *et al.* [133] framed WSCD as knowledge distillation from a Siamese teacher to a student network, with Multiscale Sigmoid Inference (MSI) post-processing refinement: evaluated on LEVIR-CD, WHU-CD, and DSIFN-CD. Liu *et al.* [134] introduced MS-Former exploiting patch-level labels through a memory-supported transformer with bidirectional attention block (BAB) for prototype-based temporal difference enhancement. Bandara *et al.* [135] presented a unified framework accommodating unsupervised, weakly supervised, regional supervised, and fully supervised change detection within a single GAN-based architecture. Wu *et al.* [136] proposed SSCMNet combining multiscale self-supervised constraints and change-mask guidance. Xie *et al.* [137] introduced ACWCD exploiting inherent transformer attention maps via Attention Refinement (AR) and a Change Priori (CP) constraint. Pei *et al.* [138] proposed Rcnet using reliable co-training between two networks with different initialisation. Zhang *et al.* [139] introduced TransWCD, a scene-adaptive joint constrained framework operating under image-level supervision.

8 Change Detection in Other Sensor Modalities

SAR. Gong *et al.* [140] combined wavelet-based fusion of log-ratio and mean-ratio difference images with RFLICM clustering, incorporating a local coefficient of variation model for speckle noise. The deep learning extension [141] applied deep belief networks for unsupervised feature pre-training.

Hyperspectral. IR-MAD [29] provides the standard classical baseline. GETNET [142] established the deep learning baseline through 2D CNN processing of mixed-affinity matrices. MSDFFN [143] achieves state-of-the-art through multi-scale difference feature representation.

Heterogeneous multi-modal. FD-MCD [117] demonstrates modality-agnostic Fourier-domain structural comparison across SAR-optical pairs. SCCN [144] maps both modalities to a shared feature space through symmetric convolutional coupling. Code-aligned autoencoders [145] enforce latent space alignment via cross-modal affinity matrices.

3D and LiDAR. Girardeau-Montaut *et al.* [146] established cloud-to-cloud (C2C) and cloud-to-mesh (C2M) distance comparison paradigms. Stilla and Xu [147] review 3D change detection across five urban application domains. Siamese KPConv [148] operates on raw point clouds achieving up to 80% mIoU on multi-class urban change detection, with pre-training on simulated data reducing annotation requirements by over 3,000×.

9 Quantitative Benchmark Comparison

Tables 2–4 present quantitative performance comparisons on LEVIR-CD+, WHU-CD, and CDD respectively. Figure 6 traces the LEVIR-CD+ performance trajectory by architectural paradigm.

Table 2: Performance on LEVIR-CD+ (256 × 256 patches, 7120/1024/2048 train/val/test). Best results in bold.

Method	**Paradigm**	**Prec.**	**Rec.**	**F1**	**OA**	**IoU**
FC-EF [68]	CNN (early fusion)	69.12	71.77	70.42	97.54	54.34
SNUNet [70]	CNN Siamese	71.07	78.73	74.70	97.83	59.62
BIT [98]	CNN+Transformer	83.91	81.20	82.53	98.60	70.26
ChangeFormer [102]	Transformer Siamese	81.34	79.97	80.65	98.44	67.58
AGCDetNet [75]	CNN (early fusion)	92.12	89.45	90.76	–	83.09
HANet [79]	CNN Siamese	91.21	89.36	90.28	99.02	82.27
CGNet [80]	CNN Siamese	93.15	90.90	92.01	–	85.21
DESSN [81]	CNN Siamese	90.99	91.73	91.36	–	–
USSFC-Net [82]	Lightweight CNN	89.70	92.42	91.04	–	–
MTCNet [99]	CNN+Transformer	90.85	89.62	90.24	97.02	82.22
SwinSUNet [122]	Pure Transformer	85.34	85.85	85.60	98.92	74.82
MambaBCD-Tiny [125]	Mamba/SSM	88.82	87.26	88.04	99.03	78.63
MambaBCD-Base [125]	**Mamba/SSM**	**89.24**	**87.57**	**88.39**	**99.06**	**79.20**

The data in Tables 2–4 reveal a consistent architectural progression on LEVIR-CD+: FCN baselines achieve F1 ≈ 70–75%; attention-augmented CNN Siamese methods reach 88–92%; CNN-Transformer hybrids attain 82–90%; the pure Swin Transformer reaches 85.60%; and the Mamba SSM achieves 88.39%. Two notable observations temper a purely linear interpretation. First, attention-augmented CNN Siamese methods (CGNet 92.01%, USSFC-Net 91.04%) outperform the pure transformer SwinSUNet (85.60%) on LEVIR-CD+, suggesting that the global attention advantage is partially offset by lower data

Table 3: Performance on WHU-CD. Best results in bold.

Method	Paradigm	Prec.	Rec.	F1	OA	IoU
FC-EF [68]	CNN (early fusion)	83.50	86.33	84.89	98.87	73.74
SNUNet [70]	CNN Siamese	88.04	87.36	87.70	99.10	78.09
BIT [98]	CNN+Transformer	89.83	90.24	90.04	99.27	81.88
HANet [79]	CNN Siamese	88.30	88.01	88.16	99.16	78.82
SwinSUNet [122]	Pure Transformer	94.08	92.03	93.04	99.50	87.00
MambaBCD-Small [125]	Mamba/SSM	95.90	92.29	94.06	99.57	88.79
MambaBCD-Base [125]	**Mamba/SSM**	**96.18**	**92.23**	**94.19**	**99.58**	**89.02**

Table 4: Performance on CDD. Best results in bold.

Method	Paradigm	Prec.	Rec.	F1
FC-EF [68]	CNN (early fusion)	83.45	98.47	90.34
UNet++ [69]	CNN	89.54	87.11	87.56
BIT [98]	CNN+Transformer	96.07	93.49	94.76
DESSN [81]	CNN Siamese	95.56	93.50	94.53
USSFC-Net [82]	Lightweight CNN	93.45	96.08	94.74
SNUNet [70]	**CNN Siamese**	**98.09**	**97.42**	**97.75**

efficiency of pure transformer training on moderate-size datasets. Second, the early-fusion AGCDetNet achieves 90.76% F1 on LEVIR-CD+, competitive with many Siamese middle-fusion architectures, demonstrating that the architectural fusion strategy is not the sole determinant of performance. On WHU-CD, the same ordering holds with absolute values 5–8% higher. CDD approaches saturation with SNUNet at 97.75% F1.

10 Open Issues and Research Directions

10.1 Data-Related Challenges

Absence of a universal benchmark. The three dominant benchmarks over-represent building change in East Asian and North American urban contexts, while the diverse geographic and ecological coverage of the statistical LULC papers reviewed in Sections 5–6 demonstrates the breadth of scenarios a universal benchmark would need to accommodate. Constructing such a benchmark would require: diverse geographic regions and ecosystems; multiple sensor modalities and resolutions; multiple change type categories (building, forest, agricultural, wetland, disaster); and multi-temporal sequences rather than single bi-temporal pairs.

Class imbalance. Changed pixels comprise 4.26–4.65% of WHU-CD and LEVIR-CD. HANet's PFBS [79] addresses this explicitly, but the majority of papers simply apply focal loss or weighted cross-entropy without systematic ablation of the imbalance mitigation

strategy.

Dataset opacity. 39 of the 100 main corpus papers use custom, unreleased datasets, preventing replication.

10.2 Methodological Limitations

Benchmarking inconsistency. Different papers using the same nominal dataset differ in train/test split strategy, patch extraction parameters, and data augmentation, rendering the numerical results in Tables 2–4 as approximate rather than exact comparisons.

Computational cost. Transformer and Mamba architectures impose substantial training and inference costs, motivating the lightweight architecture track (USSFC-Net: 1.52M parameters; LSNET; LightCDNet; TINYCD) which requires further development to consistently match full-scale model accuracy.

Multi-temporal generalisation. The vast majority of reviewed methods address only the bi-temporal pair case. Operational satellite monitoring increasingly provides dense multi-temporal stacks (as addressed by Mou *et al.* [71], Sefrin *et al.* [114], and Yin *et al.* [115]), requiring architectures that natively process arbitrary-length sequences.

10.3 Promising Research Directions

Foundation model integration. SAM integration in CS-WSCDNet [129] and diffusion pre-training in DDPM-CD [126] demonstrate clear pathways to annotation efficiency. Geospatial foundation models pre-trained on harmonised Landsat-Sentinel data (Prithvi-100M, SatMAE) offer domain-appropriate pre-training that could substantially improve cross-domain generalisation, particularly for the rare change types and novel geographic contexts where per-task supervised training datasets are unavailable.

Weakly supervised learning. The 11-paper WSCD corpus demonstrates sustained methodological progress. The image-level annotation regime substantially reduces the per-dataset labelling cost while remaining far above zero-shot capability, making WSCD particularly attractive for operational deployment on new geographic domains.

Uncertainty quantification. Operational change detection systems require calibrated uncertainty estimates for prioritising sites requiring field verification. Bayesian deep learning, conformal prediction, and MC-Dropout approaches remain underexplored in the change detection context.

Multi-modal fusion. FD-MCD [117] results on five heterogeneous sensor pairs demonstrate the feasibility of modality-agnostic structural comparison in the unsupervised setting, but the gap to same-sensor performance under supervised learning remains substantial.

11 Conclusion

This survey has provided a systematic review of change detection methods for remote sensing imagery, covering 140 non-redundant papers compiled from two complementary source corpora and spanning the full methodological arc from classical statistical operators through machine learning to the latest deep learning architectures. The analysis reveals a field that has undergone dramatic performance improvements driven by successive architectural innovations.

On the LEVIR-CD+ benchmark, the performance trajectory spans approximately 22 percentage points in F1 score between the FCN baseline of 2018 (70.42%) and the Mamba state of the art of 2024 (88.39%), achieved through a sequence of architectural innovations: FCN/Siamese-CNN, attention-augmented Siamese, CNN-Transformer hybrid, pure Transformer, and Mamba SSM. The concurrent development of weakly supervised approaches, progressing from the foundational work of [128] through the CAM-SAM integration of [129] and the knowledge distillation framework of [133], has substantially reduced annotation requirements while maintaining competitive performance.

Persistent challenges constrain the practical impact of the field. The absence of a universal benchmark, the proliferation of inconsistent evaluation protocols, and the geographic concentration of evaluation data in East Asian and North American urban settings limit the interpretability and generalisability of reported results. The most promising near-term research directions lie at the intersection of change detection with large-scale pre-training, including adaptation of geospatial foundation models, diffusion-based feature learning, and vision-language models, which promise to substantially reduce annotation requirements while improving cross-domain generalisation.

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